

Project Report - Prompt Evaluation Studio Pro

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Executive Summary

Prompt Evaluation Studio Pro is a comprehensive Prompt Engineering platform developed using Python 3.12, Streamlit, and Google Gemini API. The application provides twelve integrated modules for the complete prompt lifecycle — from creation and testing to evaluation, optimization, and analytics.

1. Introduction

1.1 Background

Large Language Models (LLMs) require carefully crafted prompts to produce reliable outputs. However, most developers lack systematic tools for prompt development, testing, and quality assurance.

1.2 Objectives

- Build a production-quality prompt engineering platform
- Demonstrate advanced prompt engineering techniques
- Implement proper software architecture with testing
- Create portfolio-ready documentation and materials

1.3 Scope

The project covers user authentication, prompt management, AI integration, evaluation, analytics, export, and deployment — excluding resume analysis (separate project).

2. System Design

2.1 Architecture

Three-tier layered architecture:

- **Presentation:** Streamlit frontend with 11 page modules
- **Business Logic:** 10 service classes via ServiceContainer
- **Data:** SQLite with SQLAlchemy ORM

2.2 Technology Selection

Component	Choice	Rationale
Language	Python 3.12	AI ecosystem standard

UI	Streamlit	Rapid AI app development
LLM	Google Gemini	Free tier, strong performance
Database	SQLite	Zero-config for portfolio
Validation	Pydantic v2	Type-safe schemas
Charts	Plotly	Interactive analytics
Testing	pytest	Industry standard
Deployment	Docker	Reproducible environments

2.3 Database Schema

Seven tables with normalized relationships. See DATABASE.md for full schema.

3. Feature Implementation

3.1 Authentication Module

- bcrypt password hashing with unique salts
- Session management via Streamlit session_state
- Remember-me token stored in database

3.2 Prompt Playground

- Full generation parameter control
- Variable substitution with {{name}} syntax
- Streaming and JSON mode support
- Auto-save experiments on every run

3.3 Comparison Engine

- Parallel execution of 2-5 prompt variants
- Metrics: time, length, tokens, quality scores
- Weighted overall scoring with winner badge

3.4 Evaluation Engine

- Nine criteria scored 0-100
- Structured JSON prompt design
- Explanation, suggestion, and optimized prompt generation

3.5 Optimizer

- Multi-version output (1-5 versions)
- Different optimization strategies
- Documented improvement explanations

4. Prompt Engineering Methodology

4.1 Techniques Applied

1. Role Prompting — Expert personas in system prompts
2. Structured JSON — Parseable evaluation outputs
3. Chain of Thought — Internal reasoning for optimization
4. Few-Shot Examples — Built-in templates
5. Prompt Chaining — Playground → Evaluate → Optimize loop
6. Constraint Specification — Explicit optimization goals
7. Output Validation — Pydantic + JSON parsing with fallback

4.2 Evaluation Criteria

Accuracy, Completeness, Hallucination Risk, Grammar, Structure, Professionalism, Formatting, Readability, Prompt Effectiveness — each scored 0-100 with weighted overall score.

5. Testing

5.1 Test Coverage

- 23 pytest tests across 6 test files
- Unit tests for utilities and metrics
- Database tests for auth and CRUD
- Mocked Gemini API tests
- Integration tests for cross-service workflows
- Startup tests for import verification

5.2 Results

All 23 tests pass successfully.

6. Error Handling

Custom exception hierarchy handles:

- Invalid API Key (INVALID_API_KEY)
- No Internet (NO_INTERNET)
- Rate Limit (RATE_LIMIT)
- Empty Prompt (EMPTY_PROMPT)
- Invalid Response (INVALID_RESPONSE)
- JSON Parse Errors (JSON_PARSE_ERROR)
- Database Errors (DATABASE_ERROR)
- Timeout (TIMEOUT)

7. Deployment

- Local: streamlit run app.py
- Docker: docker-compose up --build
- Cloud: Streamlit Cloud with secrets configuration

8. Documentation Deliverables

README, ARCHITECTURE, DATABASE, API, INSTALLATION, DEPLOYMENT, TESTING, INTERVIEW_GUIDE, PROMPT_ENGINEERING_GUIDE, CHANGELOG, portfolio_content, PROJECT_REPORT, PPT content, diagrams, screenshot guide, video script.

9. Conclusion

Prompt Evaluation Studio Pro successfully demonstrates production-quality prompt engineering platform development. The project showcases modular architecture, systematic evaluation methodology, comprehensive testing, and professional documentation suitable for fresher portfolio presentations and technical interviews.

10. References

- Google AI Gemini Documentation
- Streamlit Documentation
- SQLAlchemy 2.0 Documentation
- Pydantic v2 Documentation

****Project Version:** 1.0.0**

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